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$$\text{Opgave: } z^2 - 2z + 1 - zi = 0$$

$$\Rightarrow \Delta = (-2)^2 - 4 \cdot 1 \cdot (1 - 2i) = 4 - (4 - 8i) = (4 + 0i) + (-4 + 8i) = 8i$$

$$\Rightarrow 8i = (x + yi)^2 = x^2 - y^2 + 2xyi$$

$$\Rightarrow \begin{cases} x^2 + (-y^2) = 0 \\ 2xy = 8 \end{cases}$$

$$\Rightarrow \begin{cases} x^2 + (-y^2) = 0 \\ x^2(-y^2) = -4^2 = -16 \end{cases}$$

$$\lambda^2 - 0 \cdot \lambda - 16 = 0$$

$$\Delta = -4 \cdot (-16) = 64 \Rightarrow \sqrt{\Delta} = \pm 4$$

$$\Rightarrow \lambda_{1,2} = \frac{a \pm \sqrt{\Delta}}{2} = \pm \frac{8}{2} = \pm 4$$

$$\Rightarrow x^2 = \lambda_1 = 4 \Rightarrow x = \pm 2$$

$$\Rightarrow -y^2 = \lambda_2 = 4 \Rightarrow y = \sqrt{-4} = \sqrt{-1} \cdot \sqrt{4} = \pm 2i$$

$$x = 2 \text{ en } y = 2i \text{ of } x = -2 \text{ en } y = -2i \Rightarrow k = (2 + 2i)$$

$$z_1 = \frac{-b + k}{2a} = \frac{2 + (2 + 2i)}{2} = \frac{4 + 2i}{2} = 2 + i$$

$$z_2 = \frac{-b - k}{2a} = \frac{2 - (2 + 2i)}{2} = \frac{-2i}{2} = -i$$

References